

Deliverable 3: Contextual Embedding Pipeline & Baseline Transformer

Github repo: <https://github.com/FahadAftab14/Phishing-Detection>

1. Contextual Embedding Module (Task 1)

Goal: generate fixed contextual embeddings from email subject + body using a pretrained transformer, for later use in the fusion architecture.

- Found subject field was missing from the final feature table (dropped during the original Bucket A/C unification step). Fixed by re-merging subject from the raw Enron/Nazario files by email_id.
- Combined subject + body into one input string per email, format: [SUBJECT] ... [BODY] ...
- Checked token length distribution before choosing max_length. Median ~218 tokens, but heavy right skew (max 69,496, likely long forwarded threads). Set max_length = 512 (model ceiling) to balance coverage vs compute cost.
- Used DistilBERT-base-uncased (frozen, no fine-tuning at this stage) to generate mean-pooled embeddings, run in FP16 on free-tier T4.
- Processed all 253,229 rows in resumable, checkpointed chunks (5,000 rows/chunk) to protect against Colab session disconnects. Train: 15.2 min, Test: 4.1 min.
- Output: consolidated embedding arrays (train/test), 768-dim per email, saved to Drive and indexed by email_id. No NaNs, row order verified against source data.

2. Baseline Transformer Fine-Tuning (Task 2)

Goal: fine-tune a transformer end-to-end for phishing text classification, distinct from the frozen embeddings above.

- Class imbalance (~84:1 benign:phishing) handled via weighted cross-entropy loss rather than undersampling, to preserve all real Enron data.
- Ran into an out-of-memory crash tokenizing the full dataset in one pass. Fixed by switching to HuggingFace's datasets library with batched, disk-backed tokenization instead of holding everything in RAM at once.
- Hit an unrelated environment bug: datasets' torch-format conversion crashed on a broken torchvision import in this Colab build. Fixed by skipping that formatter and letting the data collator handle tensor conversion instead.
- Trained DistilBERT-base, 1 epoch, effective batch size 32 (4 x 8 grad accumulation), max_length 256, FP16, checkpointed every 2,000 steps. Runtime: ~35 minutes.

3. Results

Test set: 50,646 emails (50,050 benign / 596 phishing), default 0.5 threshold.

- Precision: 0.9635 | Recall: 0.8423 | F1: 0.8988
- ROC-AUC: 0.9795 | PR-AUC: 0.9121
- Confusion matrix: 502/596 phishing caught, 94 missed, 19 false positives out of 50,050 benign.

Threshold sweep showed recall can be pushed to 0.852 (508/596 caught) at threshold 0.10, at the cost of 7 additional false positives (19 to 26). No single threshold is finalized yet -- flagged as a decision to revisit once the cost of a missed phishing email vs a false positive is defined for the system's actual use case.

4. Error Analysis Finding

Missed phishing emails (false negatives) sit on domains roughly 8x older (median 1,529 vs 183 days) and with 2.5x higher reputation scores (mean 51.97 vs 20.65) than the phishing emails the model catches. IP reputation showed no difference between the two groups.

The text-only model has no access to domain data, so this reflects a real pattern: phishing hosted on older, more legitimate-looking domains also tends to read more convincingly, and a text-only classifier is blind to that correlation. This directly confirms the domain-age risk identified earlier in the literature review (Interisle data) and is direct empirical support for why the planned multimodal fusion (adding real domain reputation as an explicit input) is necessary rather than optional.